

High energy laser weapons: technology overview

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ABSTRACT

High energy laser (HEL) weapons are ready for some of today's most challenging military applications. For example, the Airborne Laser (ABL) program is designed to defend against Theater Ballistic Missiles in a tactical war scenario. Similarly, the Tactical High Energy Laser (THEL) program is currently testing a laser to defend against rockets and other tactical weapons. The Space Based Laser (SBL), Advanced Tactical Laser (ATL) and Large Aircraft Infrared Countermeasures (LAIRCM) programs promise even greater applications for laser weapons. This technology overview addresses both strategic and tactical roles for HEL weapons on the modern battlefield and examines current technology limited performance of weapon systems components, including various laser device types, beam control systems, atmospheric propagation, and target lethality issues. The characteristics, history, basic hardware, and fundamental performance of chemical lasers, solid state lasers and free electron lasers are summarized and compared. The elements of beam control, including the primary aperture, fast steering mirror, deformable mirrors, wavefront sensors, beacons and illuminators will be discussed with an emphasis on typical and required performance parameters. The effects of diffraction, atmospheric absorption, scattering, turbulence and thermal blooming phenomenon on irradiance at the target are described. Finally, lethality criteria and measures of weapon effectiveness are addressed. The primary purpose of the presentation is to define terminology, establish key performance parameters, and summarize technology capabilities.

Keywords: High Energy Laser weapons, chemical lasers, solid state lasers, free electron laser, beam control, adaptive optics, atmospheric propagation, target lethality, missions, scenarios.

1. LASER FUNDAMENTALS

The laser, an acronym for light amplification by stimulated emission of radiation, is a source of nearly monochromatic, coherent, electromagnetic radiation that propagates as a beam. The light from a laser differs in several significant ways from the light from an ordinary incandescent bulb. The electromagnetic waves emitted by an ordinary source have many different wavelengths, or colors, and often appears "white". Furthermore, these waves are uncoordinated in that their crests and valleys exhibit no well defined pattern with respect to each other. Finally, emission from individual points from within the ordinary source propagates in all directions. In contrast, laser light possesses a well defined color (monochromatic), the waves are coordinated with crests and valleys in phase with each other (coherent), and the light propagates in a well defined direction (as a beam).

The physical process by which laser light is generated is termed stimulated emission and was discovered by Albert Einstein in 1916. Light can interact with matter through several processes, including absorption, spontaneous emission, and stimulated emission as shown in Figure 1. Consider a material with several states of quantized internal energy, E_0 , E_1 , and E_2 . In thermal equilibrium, most of the atoms or molecules are in the state of lowest energy. This situation is illustrated with a large number of open circles populating the energy level, E_0 . When light is incident on such a material, as illustrated by the wave packet, the intensity may be attenuated as it is transmitted through the material via the process of absorption. The energy associated with the attenuated light is deposited into the material and some of the atoms or molecules transition to a state of higher energy, E_2 . The particle undergoing this transition is shaded grey in the initial state and black in the final state of Figure 1.

The light emitted from an ordinary source is generated by spontaneous emission. If an atom or molecule is in an excited state, E_2 , one of the ways it can decay to a state of lower energy, E_1 , is by emission of a photon. The photon is a packet

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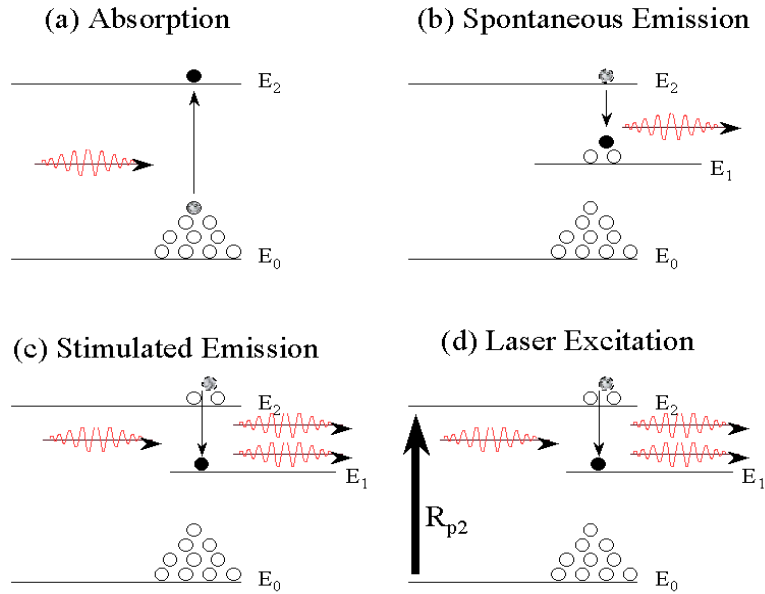


Figure 1. The interaction of light with matter via: (a) absorption, (b) spontaneous emission, and (c) stimulated emission. Pumping of an excited state to produce a population inversion is shown in (d).

of light with energy equal to the difference in energy between the emitting and final states. The frequency of the emitted light, $\nu = (E_2 - E_1) / h$, is related to the energy difference, $E_2 - E_1$, by Planck's constant, $h = 6.6256 \times 10^{-34}$ J-s. The frequency and wavelength, λ , of light propagating in a medium with index of refraction, $n(\lambda)$, are related by the speed of light, $c = 2.9979 \times 10^8$ m/s:

$$\lambda \nu = c / n(\lambda) \quad (1)$$

Spontaneous emission occurs without external influence and is independent of other emission from within the sample.

Alternatively, emission may be stimulated by an incident photon. In this stimulated emission process, the newly emitted photon possesses the same energy, direction of propagation, and phase as the incident photon. An exact copy of the incident photon is generated and the intensity of the light is amplified as the wave continues to propagate through the medium. This is the mechanism by which laser light is generated and is illustrated in Figure 1c.

Both stimulated emission and absorption will occur between a pair of energy levels, yielding an intensity variation, $I(z)$, as the wave propagates a distance z of:

$$I(z) = I(z=0) e^{\sigma(N_2 - N_1)z} \quad (2)$$

where N_1 and N_2 are the number of atoms or molecules per unit volume with energy E_1 and E_2 , respectively. The laser gain, γ , is specified by the product of the cross-section for stimulated emission, σ , and the population inversion, $\Delta = (N_2 - N_1)$

$$\gamma = \sigma \Delta \quad (3)$$

To achieve net amplification, $\gamma > 0$, and thus lasing, a population inversion, $N_2 > N_1$, is required. Normally, there are more atoms and molecules in the lower energy state and the rate of absorption is greater than the rate of stimulated emission. However, if there are more atoms or molecules in the state of higher energy, then stimulated emission will dominate and amplification of the incident light may be achieved. Thus, a population inversion between the two energy levels is

required to produce a laser. This is a situation not normally found in nature. The laser medium must be "pumped" by an external energy source to selectively deposit energy in the upper laser level, E_2 . A chemical laser achieves the population inversion from the products of a chemical reaction.

When the gain exceeds the total optical losses, α , due to mirror output coupling, scattering losses, and resonator losses, then net amplification and lasing is achieved. The threshold population inversion is defined as the inversion required for the gain to equal the losses:

$$\Delta_{th} = \gamma_{th} / \sigma = \alpha / \sigma \quad (4)$$

Both stimulated emission and absorption will occur between a pair of energy levels. Normally, there are more atoms and molecules in the lower energy state, and the rate for absorption is greater than stimulated emission. Under these circumstances, light is attenuated as it propagates through the medium. However, if there are more atoms or molecules in the level of higher energy, then stimulated emission will dominate and amplification of the incident light may be achieved. Thus, a population inversion between the two energy levels is required to produce a laser. To achieve this population inversion, the medium must be pumped externally to selectively deposit energy in the upper laser level, E_2 . This laser pumping, or excitation, is shown schematically in the last segment of Figure 1.

It was not until 1960 that T.H. Maiman used these ideas and demonstrated the first laser device, the ruby laser. A laser typically consists of an energy source, a gain medium, and an optical resonator, as shown in Figure 2. The energy source is required to disturb the sample of atoms or molecules from equilibrium and pump the upper laser level. The gain medium provides for the amplification of light, which initially arises from spontaneous emission from within the sample. The optical resonator is usually a set of mirrors which provide feedback of light for multiple passes through the gain medium to establish an intense beam of radiation. Considering the light within the optical resonator as an electromagnetic wave, one can see that the wave must replicate its phase upon a round trip for the two wave trains to constructively interfere. Thus, the optical resonator may be designed to select a single mode and produce nearly monochromatic radiation.

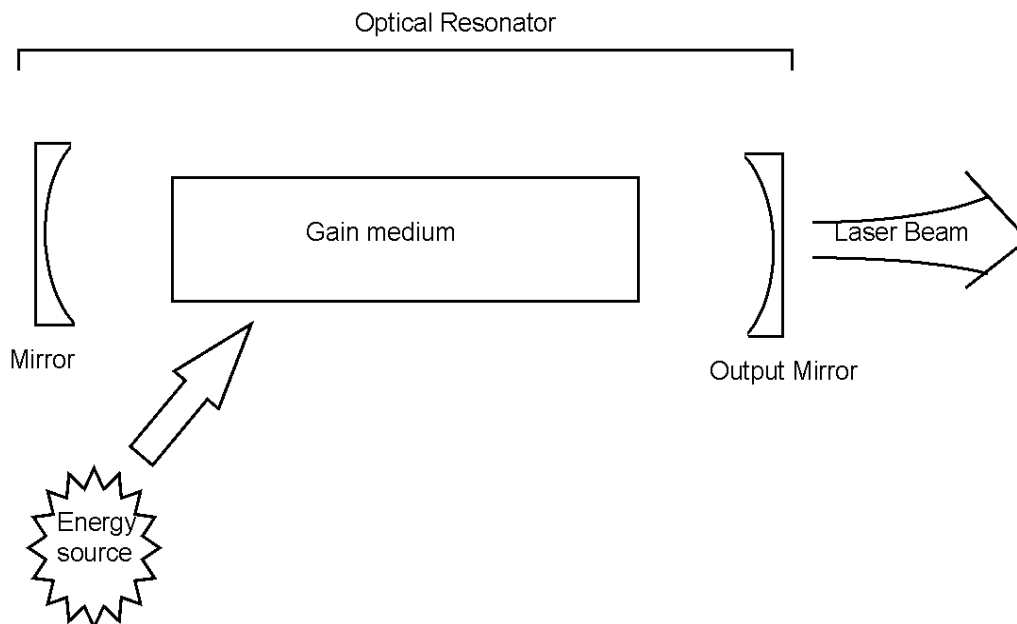


Figure 2. A laser is typically composed of three major components: a gain medium where the light is amplified, an external energy source to produce the population inversion, and an optical resonator or laser cavity which provides feedback to the gain medium and establishes a well defined laser beam.

The development of the laser represents a significant scientific and technical achievement. In 1964 the Nobel Prize in physics was awarded to C.H. Townes, N.G. Basov, and A.M. Prokhorov for their fundamental work in quantum electronics, particularly leading to the construction of the laser.

3. CHARACTERISTICS OF LASER WEAPONS

After decades of research and development, high energy laser weapons are becoming an operational reality. High energy laser (HEL) weapons are being developed and tested by all branches of the US military and have the potential to revolutionize modern warfare. The transition of these laser weapons to the battlefield has begun with programs such as the Airborne Laser, Tactical High Energy Laser, and Advanced Tactical Laser. In April 1997, Secretary of the Air Force Widnall recognized: "It isn't very often an innovation comes along that revolutionizes our operational concepts, tactics and strategies. You can probably name them on one hand - the atomic bomb, the satellite, the jet engine, stealth, and the microchip. It's possible the airborne laser is in this league."¹

Laser weapons offer a number of advantages over conventional weapon systems. First, the speed of light delivery enables the war-fighter to engage very distant targets immediately following detection. Debris fields may be limited to enemy territory, enemy evasion or maneuver is highly constrained, and time critical targets can be engaged. Secondly, HEL systems offer covert operation and low collateral damage. Lasers are a "surgical" weapon with unprecedented aim-point selection. Once deployed, HEL weapons offer many inexpensive engagements. However, initial deployment costs are high. Finally, lasers can provide graduated effects, tailoring the range of results to non-lethal, disruptive and destructive effects.

The elements of laser weapon scenarios include: (1) the engagement setting (ground-to-air, air-to-ground, air-to-air, etc), (2) target parameters such as damage threshold and speed, (3) platform constraints such as payload, (4) environmental aspects including the state of the atmosphere and presence of particulates, and (5) engagement parameters such as range and magazine depth. Strategic scenarios might be represented by platform payloads of up to 100,000 lbs, ranges of 1000 km, and device powers in the megawatt class.

The DoD Laser Master Plan evaluated several offensive and defensive applications for HEL weapons. Table 1 summarizes the difficulty of these applications : yellow – very difficult to acquire targets, control HEL beam, and destroy targets; green – moderately difficult; blue – less difficult.

Table 1. Application concepts for HEL weapons

Scenario	Application	Targets	Perceived Difficulty
Defensive	National Missile Defense	Ballistic Missile	Yellow
Defensive	Theater Missile Defense	Ballistic Missile	Green
Defensive	Area or Point Defense	Cruise Missiles, Rockets	Blue
Defensive	Area or Point Defense	Mortars, Artillery	Yellow
Defensive	Platform Defense	Cruise Missile	Blue
Offensive	Airborne Precision Strike	Aircraft, Surface Craft	Blue
Offensive	Non-Lethal Weapons	Aircraft, Surface Craft	Blue
Offensive	Anti-Satellite	Satellites	Green

4. HEL WEAPON SYSTEM COMPONENTS

The components of an HEL weapon system are shown in Figure 3, and include the high energy laser device, a beam director and beam control system, the acquisition, tracking and pointing system, degradation of the laser beam as it is transmitted through the atmosphere, and finally target lethality. The performance of the weapon system depends on many factors. A few of the most critical quantities are listed in the figure.

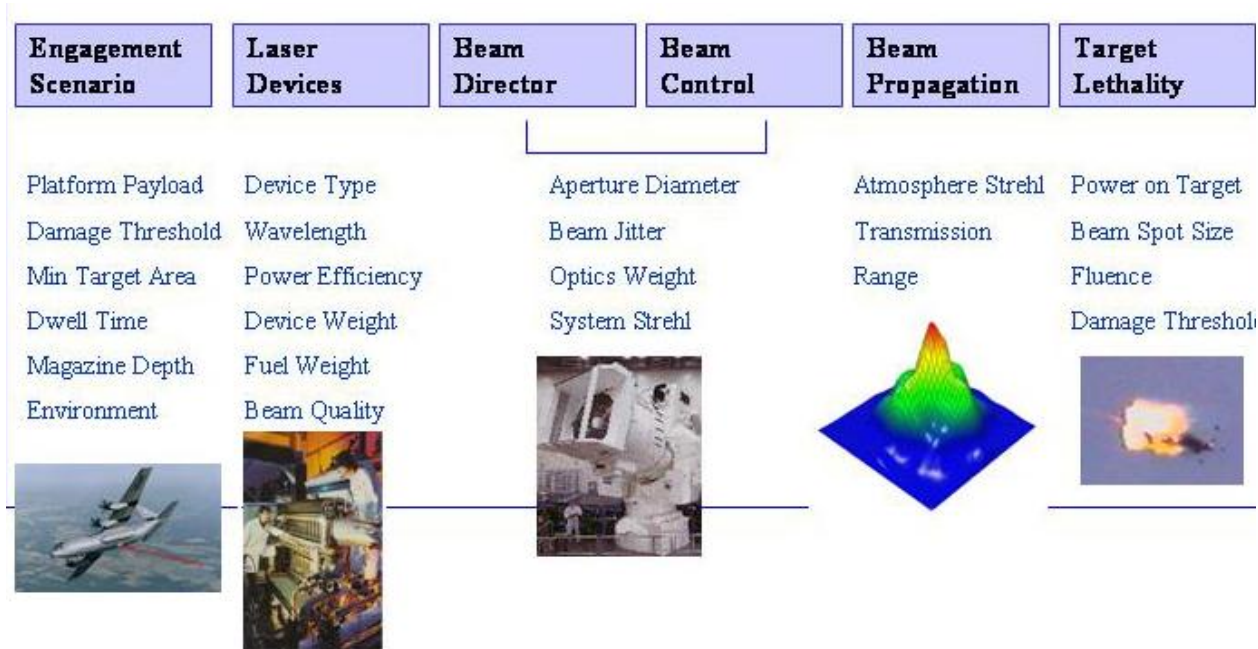


Figure 3. The components of a High Energy Laser (HEL) weapons system.

DoD currently funds three kinds of laser device technology for High Energy Laser weapons: chemical, solid-state and free electron. A chemical laser uses an exothermic reaction as an energy source to pump a low pressure gas phase gain medium. Chemical lasers are usually large, high power devices that integrate chemical delivery systems, a supersonic nozzle and an optical resonator. They require specialized fuels and exhaust waste heat. The solid-state lasers include heat capacity, fibers and continuously-cooled lasers. In the past, flashlamp-pumped solid-state lasers have produced high peak power, at the kilowatt level with good beam quality. However, it is the development of high efficiency, laser diode arrays that enables the possibility of higher average power solid-state laser weapons. They require specialized thermal management systems. A free electron laser uses a relativistic electron beam as its lasing medium. Bunches of high-energy electrons move through a vacuum while magnets cause them to wiggle and produce light. FEL's typically operate with very short pulses at high repetition rates. They have good beam quality, are widely tunable, and operate best at high power. Each of these laser devices are discussed in detail in the following sections.

4.1 Chemical Lasers

4.1.1. HF/DF Lasers

The chemical laser depends on the energy liberated from an exothermic chemical reaction to produce the population inversion. Polanyi first proposed a chemical laser in 1960 by recognizing that many chemical reactions deposit excess energy in the stretching of the newly formed bond.² If the chemical reaction favors the production of states with greater internal energy over states with less internal energy, then a population inversion can be achieved.

As an example of using a chemical reaction to pump a laser, consider the reaction of atomic fluorine with molecular hydrogen to produce vibrationally excited hydrogen fluoride, as shown in Figure 4. This reaction produces 130 kJ/mole excess energy which can be deposited in vibration of the newly formed HF bond.³ The first four vibrational levels, $v=0-3$, are accessible. The barrier to this reaction is small, 5 kJ/mole, and thus the rate for producing HF is quite rapid. The key to developing an HF laser based on this reaction is the relative rates for producing the various vibrational levels of HF. These relative rates have been measured as 12:20:6:1 for $v=3:2:1:0$.⁴ Thus, 31% of the HF products will be found in the vibrational level $v=3$. The most probable product is $v=2$ at 51%. Only 3% of the reactions yield the lowest vibrational state, $v=0$. Clearly, a population inversion between vibrational levels in HF can be generated by this direct chemical reaction.

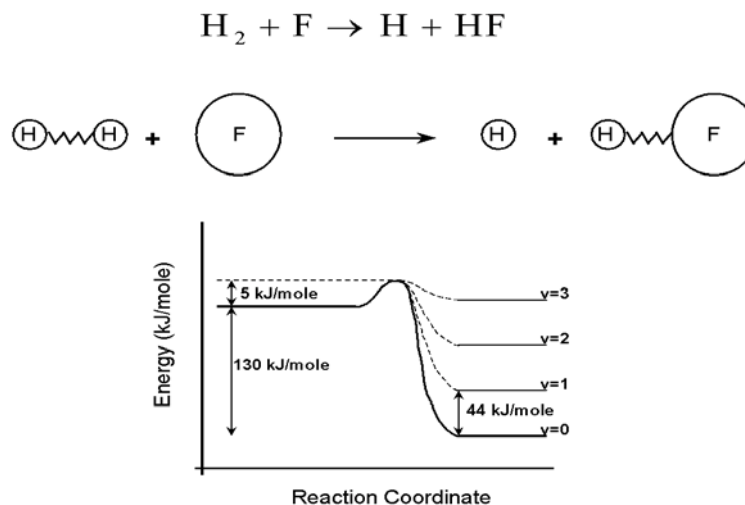


Figure 4. The reaction of atomic fluorine with molecular hydrogen produces vibrationally excited hydrogen fluoride. The reaction coordinate represents the breaking of the H_2 bond as the fluorine atom approaches and the subsequent formation of the HF bond as the product hydrogen atoms depart the collision center.

In 1965 Kasper and Pimentel demonstrated the first chemical laser by initiating a hydrogen-chlorine explosion with a flashlamp.⁴ Within a few years, purely chemical, continuous operation HF lasers based on the chemistry described above had been demonstrated and by 1984 HF lasers with powers greater than 1 MW (one million watts) were developed.⁵

The HF laser operates in the infrared region of the electromagnetic spectrum with a wavelength near $2.7 \mu\text{m}$. The output power as a function of laser wavelength is shown in the simulation of Figure 5. The HF laser can operate at several discrete wavelengths corresponding to different vibrational and rotational transitions. These transitions are labeled according to their vibrational (v) and rotational (J) quantum numbers. The earth's atmosphere is nearly opaque at the HF laser wavelengths, as seen in Figure 5. However, by substituting deuterium for hydrogen, the DF laser can be made to lase at $\lambda = 3.6 - 4.2 \mu\text{m}$. The atmosphere exhibits a window with good transmission in this range, making the DF laser preferred for applications requiring atmospheric propagation.

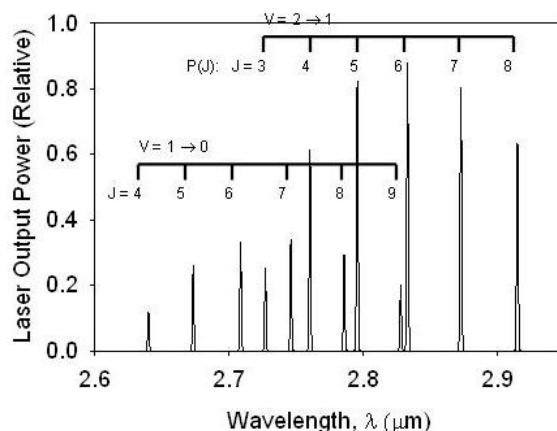


Figure 5. Simulated spectrum of an HF laser showing several discrete laser wavelengths. The transitions between ro-vibrational levels are labeled according to spectroscopic notation.

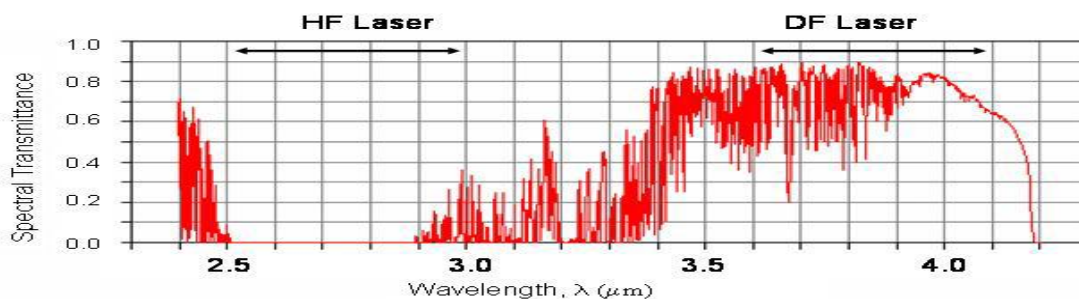


Figure 6. The atmosphere is opaque to the HF laser. However, the longer wavelength DF laser operates in an atmospheric window.

The basic components of a combustion driven, supersonic mixing HF laser are shown in Figure 7. Fluorine atoms must be produced to drive the $F + H_2 \rightarrow HF + H$ reaction and this is accomplished by thermal dissociation of molecular fluorine at high pressure and temperature in the combustor. Typically, molecular deuterium is injected as the fuel and burned with molecular fluorine or NF_3 to produce 30-50% dissociation of F_2 at 2000 - 4000 °K and 1 - 15 kPa total pressure. These temperature and pressure conditions are far too high for efficient laser operation, and the effluents of the combustor including the atomic fluorine are supersonically expanded through a nozzle assembly at Mach 1-3 producing a temperature of 300-500 °K and a pressure of 0.2 - 0.8 kPa in the gain region. While the flow is cooled, the fluorine atoms remain dissociated as there is insufficient time for recombination to molecular fluorine. Molecular

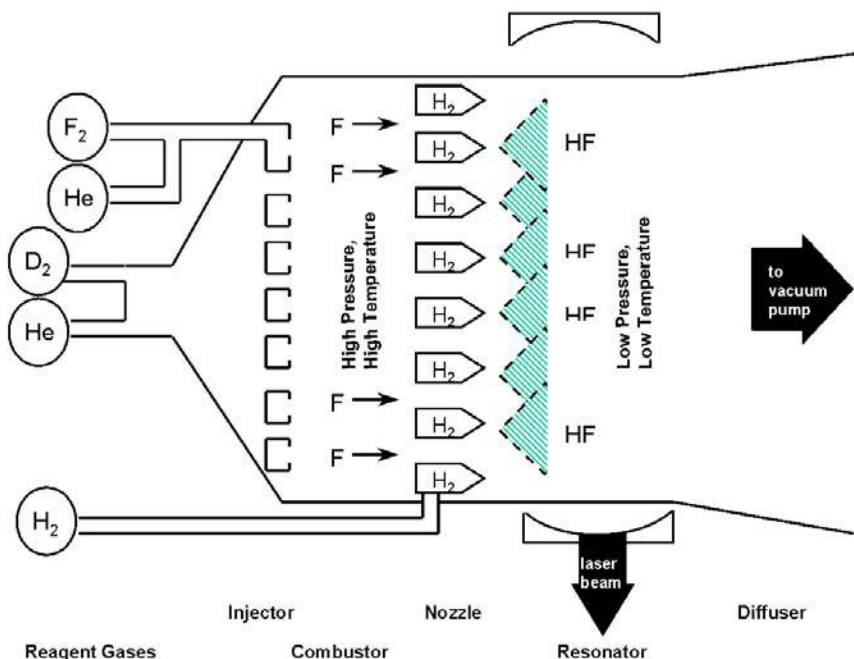


Figure 7. Schematic diagram of a typical supersonic mixing hydrogen fluoride chemical laser.

hydrogen is then injected into this supersonic expansion through a large number of very small nozzles to enable good mixing and efficient reaction with the atomic fluorine to produce the vibrationally inverted HF molecules for lasing. Another function of the supersonic flow is stretch the reaction zone for the $F + H_2 \rightarrow HF + H$ reaction over the full width of the laser resonator. The laser cavity is formed so that the laser beam propagates in a direction perpendicular to the gas flow. Finally, the diffuser is designed to recovery as much of the inlet stagnation pressure as possible before exiting to the mechanical or steam injection vacuum pumping system where the gases are exhausted.

Megawatt class power levels were first achieved by the Mid-Infrared Advanced Chemical Laser (MIRACL) in the early 1980's. The MIRACL device is a cw DF chemical laser with a linear nozzle configuration which produced a square 14 cm x 14 cm beam with outcoupled power in excess of 1 MW. A modest beam quality of two times the diffraction limit was achieved. The limit to scaling the gain length using a linear geometry is nearly reached in this device.

The Alpha cylindrical HF chemical laser is also a mega-watt class laser which wraps arrays of nozzle modules in the form of a large cylinder. The fluorine and diluent flow passes radially out around the annulus that holds the hypersonic hydrogen injectors. Curvature in the radial direction on the annular mirror allows mode control and intensity levels to be controlled in the compacted leg. The cylindrical configuration is preferred for very high powers over that of conventional linear laser configurations. This provides many times the device output power without increasing the nozzle power flux, cavity injector length, or mode width. Such a resonator reduces effects from flow-filed disturbances, diffraction losses, anomalous dispersion, and other optical loss mechanisms that tend to produce a poor-quality beam. In 1991, the Alpha laser achieved megawatt power using operating conditions consistent with the low pressure environment required for space based operation.

The Tactical High Energy Laser (THEL) uses a high power chemical laser for protection of theater-level combat forces and marks the transition of high power lasers to the battlefield. The THEL is designed to be a transportable, tactical-sized, DF laser system including the laser device, pointing and tracking system, beam director, and command and control system. In June 2000, the THEL successfully intercepted and destroyed a Katyusha rocket at the High Energy Laser Systems Test Facility (HELSTF), White Sands Missile Range, New Mexico.

4.1.2 The Chemical Oxygen-Iodine Laser (COIL)

To extend chemical laser operation to significantly shorter wavelengths than the HF/DF laser requires a system which establishes an inversion between the more energetic electronic states of an atom or molecule. There are several advantages to shorter wavelength chemical lasers including higher power per mass of reagents and reduced beam divergence due to diffraction at the limiting aperture.⁷ Rather than depositing excess reaction energy into the stretching of the newly formed bond (vibration), the chemical reaction could yield atoms or molecules with internal excitation of the electrons. For example, molecular oxygen has two excited electronic states with electronic energy of $T_e = 95$ and 158 kJ/mole above the ground electronic state,⁸ as shown in the potential energy curves of Figure 8. To understand these curves, consider the oxygen molecule as two masses connected by a spring, where the bonding electrons shared by the two oxygen atoms provide the restoring force. As the oxygen molecule vibrates, the distance between the two oxygen atoms, R , changes between the two extremes, and the potential energy curve represents this range in internuclear separation at various energies, V . At large enough vibrational energy, the internuclear separation becomes quite large and the molecule dissociates to two ground state oxygen atoms. The curve labeled $X^3\Sigma_g^+$ is the lowest electronic state of molecular oxygen, which has a dissociation energy of $D_e = 489$ kJ/mole and a bond length of $R_e = 0.1207$ nm. If the electron cloud around the oxygen molecule is excited, then the effective spring force constant is changed, and the excited electronic states, labeled $a^1\Delta_g$ and $b^1\Sigma_g^+$, are generated.

To efficiently produce an inversion between electronic states requires the chemical reaction to selectively produce a specific excited electronic state over all other states, particularly the ground electronic state. Reactions involving light atoms and molecules tend to preserve the total electronic spin. Thus, the generation of an electronic inversion is aided by a difference in electronic spin between the ground and excited electronic states. However, such transitions exhibit slow spontaneous emission rates, and thus very low stimulated emission cross-sections and insufficient gain to build a laser. For example, the $a^1\Delta - X^3\Sigma$ transition in O_2 shown in Figure 8 exhibits a particularly slow spontaneous emission rate of $2.58 \times 10^{-4} \text{ s}^{-1}$. (ref 9) However, these long-lived, or metastable, electronically excited states establish a large

energy reservoir that could be used to pump a chemical laser. By collisionally transferring the energy stored in these metastable states to a suitable lasing species, a new class of chemical laser, the metastable energy transfer chemical laser, can be developed. The Chemical Oxygen-Iodine is the prime example of such an energy transfer laser.

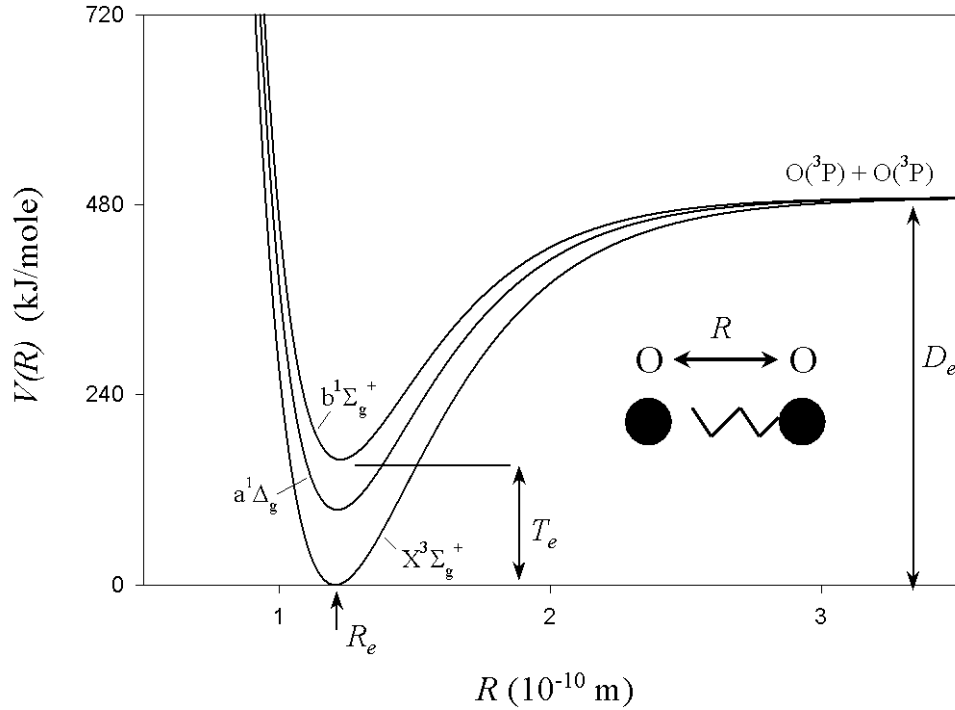
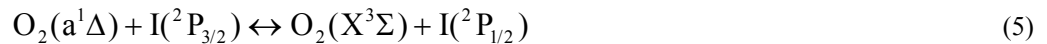


Figure 8. Morse potential energy curves for the three lowest electronic states of molecular oxygen, which dissociate to ground state oxygen atoms, $O(^3P)$. The dissociation energy, D_e , for the ground state, $X^3\Sigma_g^+$, and the electronic energy, T_e , for the second excited state, $b^1\Sigma_g^+$, are indicated.

The Chemical Oxygen-Iodine Laser (COIL) was first demonstrated at the Air Force Weapons Laboratory in 1977.¹⁰ The first electronically excited state of molecular oxygen, $O_2(a^1\Delta)$, is produced by a two phase reaction of chlorine gas and liquid basic hydrogen peroxide with near 100% efficiency. Once produced, this singlet state of oxygen can be transported for considerable distances even at moderately high pressures, as the radiative lifetime is ~ 64 minutes and $O_2(a^1\Delta)$ is very resistant to collisional deactivation. Thus, $O_2(a^1\Delta)$ is a metastable energy reservoir. Energy densities of $10\text{-}20 \text{ kJ/m}^3$ are typically attained.

The Chemical Oxygen-Iodine Laser operates on an inversion between the $5^2P_{1/2}$ and $5^2P_{3/2}$ spin-orbit split states of atomic iodine at a wavelength of $1.315 \mu\text{m}$. A near resonance exists between $O_2(a^1\Delta)$ and $I(^2P_{1/2})$ with an energy difference of only 3.3 kJ/mole , and laser pumping is achieved by collisional energy transfer from the $O_2(a^1\Delta)$ metastable energy reservoir, as shown in Figure 9. The energy transfer is rapid and a near equilibrium between the upper laser level, $I(^2P_{1/2})$, and $O_2(a^1\Delta)$ is quickly established:



The ratio of excited and ground state atomic iodine concentrations is approximately determined by the equilibrium constant for reaction (5), $K_{eq} = 0.75 \exp(402/T)$:

$$\frac{[I(^2P_{1/2})]}{[I(^2P_{3/2})]} = \frac{[O_2(a^1\Delta)]}{[O_2(X^3\Sigma)]} K_{eq}(T) \quad (6)$$

where the bracket $[]$ indicate concentration of the indicated specie.

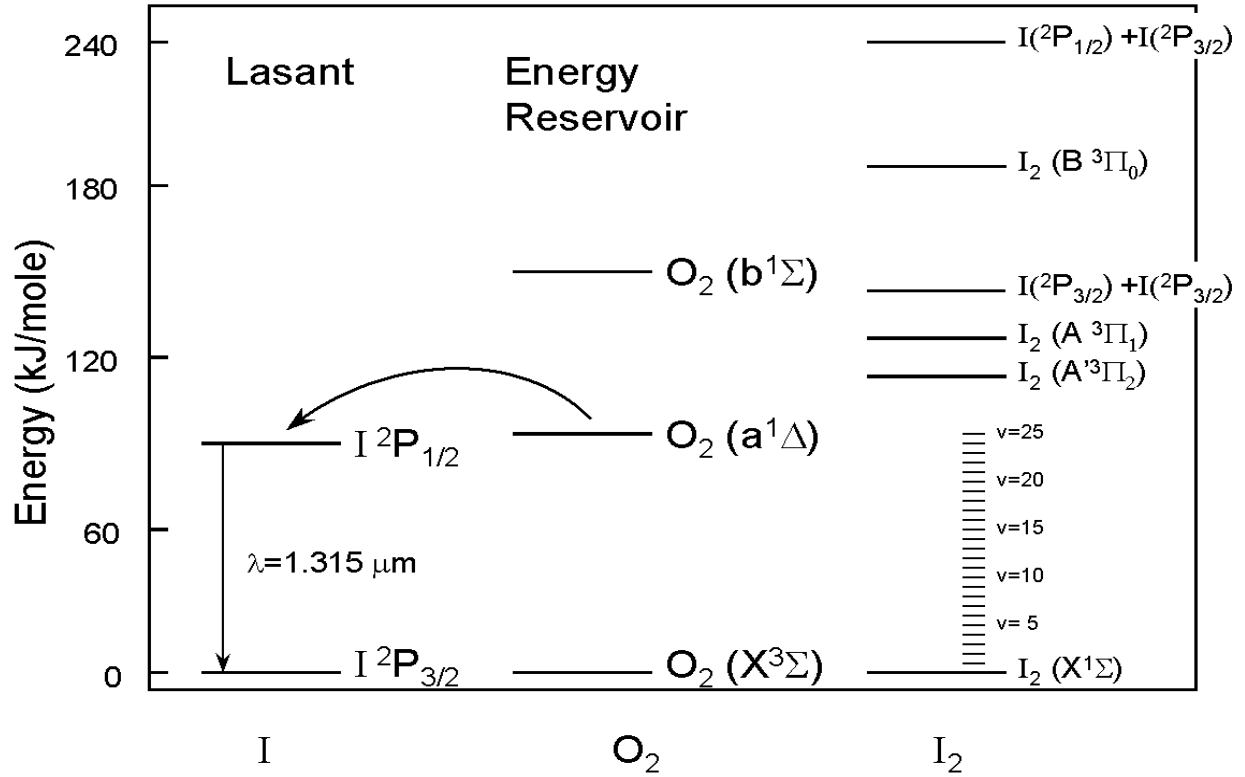


Figure 9. Energy level diagram for the Chemical Oxygen-Iodine Laser illustrating the resonant energy transfer from metastable reservoir $O_2(a^1\Delta)$ to the upper laser level of atomic iodine, $I(^2P_{1/2})$. The energy levels for several key states involved in the dissociation of molecular iodine are also provided.

The atomic iodine $5^2P_{1/2} \rightarrow 5^2P_{3/2}$ lasing transition is composed of six hyperfine components, as shown in Figure 10. The nuclear spin is $I=5/2$. The $5^2P_{1/2} (F'=3) \rightarrow 5^2P_{3/2} (F''=4)$ component provides the highest gain and lasing occurs on this individual transition. Note that the degeneracies of the hyperfine components are $g_F = (2F+1)$, so that the degeneracy of the upper and lower laser levels are $g_2=7$ and $g_1=9$, respectively. If all the hyperfine levels are populated statistically:

$$\begin{aligned} [I(^2P_{1/2}, F'=3)] &= \left(\frac{7}{12}\right) [I(^2P_{1/2})] \\ [I(^2P_{3/2}, F''=4)] &= \left(\frac{9}{24}\right) [I(^2P_{3/2})] \end{aligned} \quad (7)$$

then the population inversion is:

$$\begin{aligned} \Delta &= [I(^2P_{1/2}, F'=3)] - \left(\frac{g_2}{g_1}\right) [I(^2P_{3/2}, F''=4)] = \\ &= \left(\frac{7}{12}\right) ([I(^2P_{1/2})] - 0.5[I(^2P_{3/2})]) = \left(\frac{7}{12}\right) \left\{ K_{eq} \left(\frac{Y}{1-Y} \right) - \frac{1}{2} \right\} [I(^2P_{3/2})] \end{aligned} \quad (8)$$

where

$$Y = \text{yield} = \frac{[O_2(a^1\Delta)]}{[O_2(X^3\Sigma)] + [O_2(a^1\Delta)]} \quad (9)$$

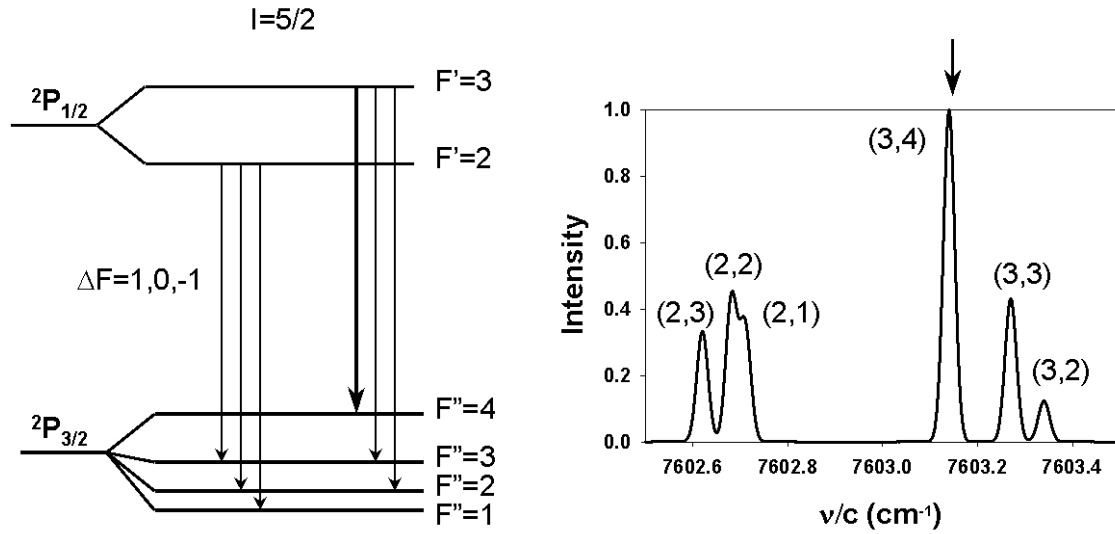
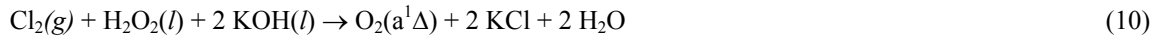


Figure 10. Hyperfine structure and spectrum of the $I(2P_{1/2} \rightarrow 2P_{3/2})$ transition in atomic iodine indicating the greatest gain is achieved for the $F'=3 \rightarrow F''=4$ component near $\nu = 7603.2 \text{ cm}^{-1}$ or $\lambda = 1.315 \text{ }\mu\text{m}$. The spectrum is simulated using lineshapes of equation (5) with a linewidth of $\Delta\nu_D = 840 \text{ MHz}$.

Threshold for lasing (positive inversion) is achieved at $T=295 \text{ K}$ when the ratio of $O_2(a^1\Delta)$ to total oxygen, or yield of singlet oxygen, is approximately 15%. The threshold is reduced considerably at lower temperatures due to the temperature dependence of the equilibrium constant. At $T=160 \text{ K}$, a temperature consistent with typical nozzle exit conditions, the threshold yield is reduced to 5%. The cross-section for stimulated emission at room temperature is $\sigma = 7.4 \times 10^{-18} \text{ cm}^2$ and the gain in a COIL device is typically near $\gamma < 1\% / \text{cm}$.¹¹

A schematic diagram of a typical supersonic COIL device is shown in Figure 11. The chemical generator of $O_2(a^1\Delta)$ is a two phase reactor governed by the stoichiometry:



The $\text{KOH} / \text{H}_2\text{O}_2$ solution is termed the basic hydrogen peroxide solution, or BHP.

The earliest singlet oxygen generators were of a sparger design, where chlorine gas is bubbled through the liquid hydrogen peroxide solution. The two phase reaction requires large liquid surface to volume ratios for efficient operation, and the second generation of chemical generators were based on a wetted wall reactor, as shown in Figure 12.¹¹ By rotating a stack of disks through a pool of basic hydrogen peroxide and past a stream of chlorine gas, the reaction is confined to a thin film on each disk. The time required to diffuse the singlet oxygen from within the liquid phase to the gas phase interface is reduced and the deactivation of $O_2(a^1\Delta)$ is minimized. Advanced generators involving sprays of droplets with near uniform size are now being used for COIL devices.

The effluent from the singlet oxygen generator includes $O_2(a^1\Delta)$ and water vapor. Water vapor is an efficient collisional quencher of the upper laser level, $I(2P_{1/2})$, and reduces device performance. Thus a water trap or condenser is

sometimes used to remove the water from the gas stream. The total pressure at the exit of the generator is typically 1.5 - 6.5 kPa and usually includes helium as a diluent.

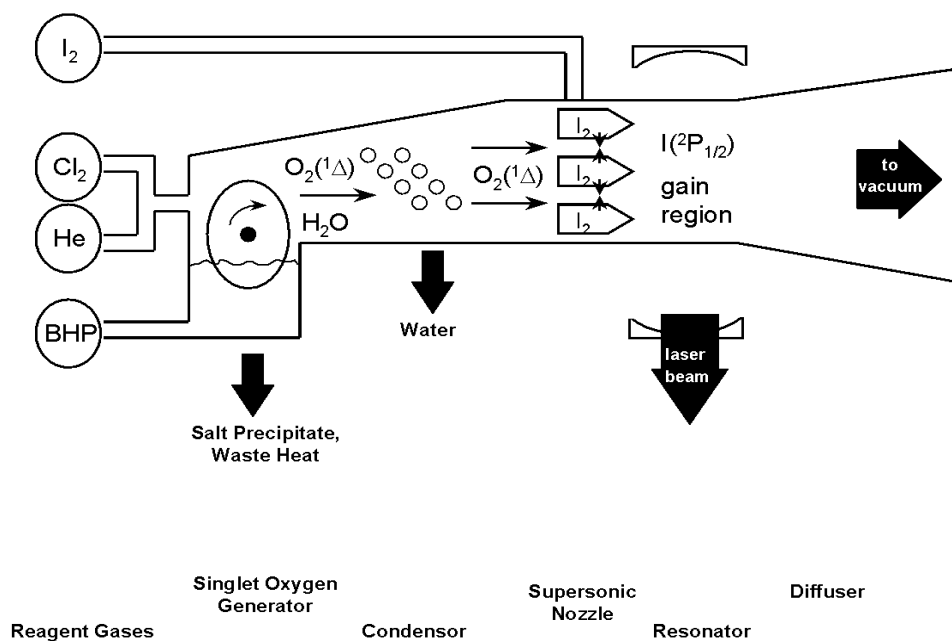


Figure 11. Schematic diagram of a typical supersonic Chemical Oxygen-Iodine Laser (COIL).

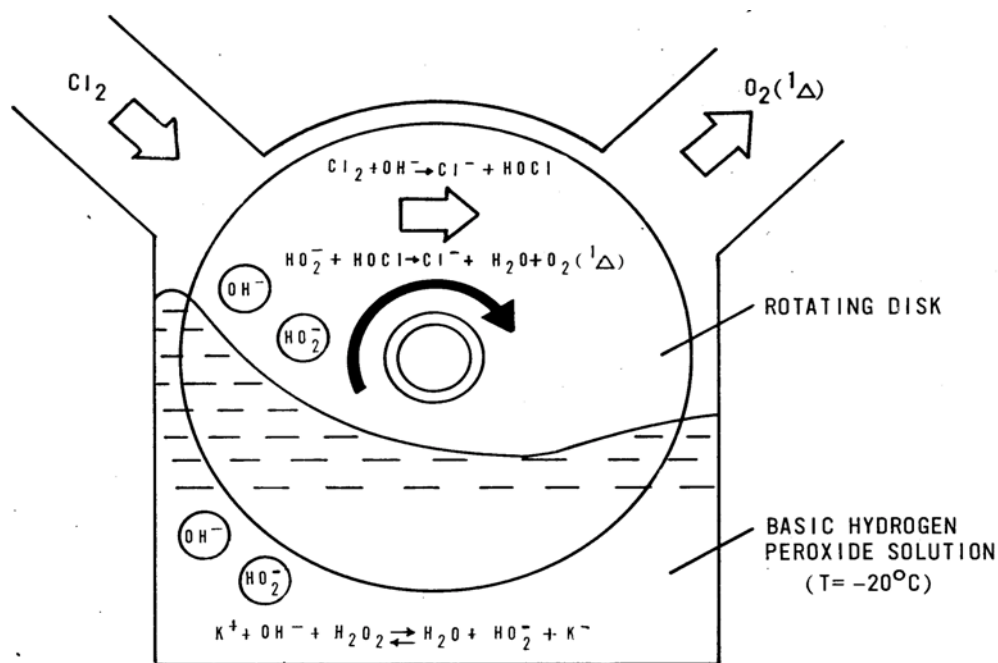


Figure 12. Rotating disk design for the chemical generation of singlet oxygen, O₂(a¹Δ), via the two phase reaction of chlorine gas with liquid hydrogen peroxide (from reference (11)).

Molecular iodine is injected transverse to the primary oxygen flow near the throat of a supersonic nozzle, as shown in Figure 13.¹¹ The molecular iodine is rapidly dissociated to atomic iodine by the presence of the singlet oxygen in a complex, multi-step mechanism so that the flow is fully dissociated as it reaches the exit plane of the nozzle and before entering the gain region.

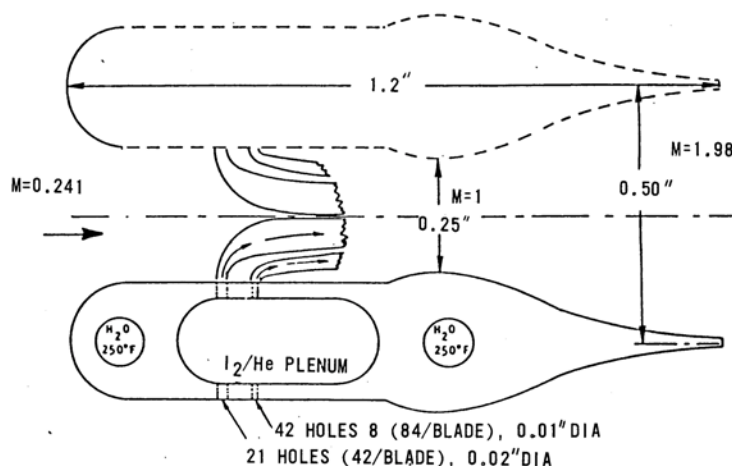


Figure 13. Typical supersonic mixing nozzle for a Chemical Oxygen-Iodine Laser where molecular iodine is injected transverse to the flow of singlet oxygen (from reference (11)).

The iodine dissociation process is poorly understood and provides the greatest difficulty in modeling the gas phase kinetics of the COIL device. The energetics of the dissociation process are shown in Figure 9. The second electronically excited state of oxygen, $O_2(b^1\Sigma)$, is produced from the energy pooling reaction:



and $O_2(b^1\Sigma)$ is sufficiently energetic to dissociate iodine. However, there is insufficient concentration of $O_2(b^1\Sigma)$ to explain the phenomenological dissociation rate, particularly when significant water vapor is present. A slow initiation of the dissociation process may begin by $O_2(a^1\Delta)$ excitation of vibrationally excited iodine, I_2^* , followed by a second $O_2(a^1\Delta)$ collision producing ground state atomic iodine. The resulting iodine atoms are rapidly excited to the spin orbit state, $I(^2P_{1/2})$, by energy transfer. The dissociation then rapidly accelerates as $I(^2P_{1/2})$ replaces $O_2(a^1\Delta)$ as the dominant partner for production of I_2^* . This dissociation process removes at least 2, and sometimes as many as 6-18, $O_2(a^1\Delta)$ molecules per dissociated I_2 molecule, depending on the rate for deactivating the intermediate, I_2^* , and any quenching of $I(^2P_{1/2})$. In order to avoid consuming a large fraction of the energy stored in the metastable reservoir in this dissociation process, the ratio of molecular iodine to singlet oxygen flow rates is usually small, about 1%. During the lasing process, the iodine atoms will experience many energy transfer excitation and stimulated emission cycles to extract most of the energy stored in $O_2(a^1\Delta)$. Recent kinetic studies have suggested alternative dissociation mechanisms.

Subsonic COIL devices were developed in the United States during the period 1977 – 1984. A rapid set of initial demonstrations took the subsonic COIL device from 4 milliwatts to 100 watts in just a few years. Scaling to 4.6 kW was achieved by 1982 with the 4-meter gain length in the COIL IV device.

An abrupt change in emphasis occurred with the first demonstration of a supersonic Chemical Oxygen-Iodine Laser in 1984. The reduction in device size, the lower cavity operating temperature and the longer gain lengths were immediately recognized as critical advantages. The ReCOIL device, the first supersonic COIL, boasted 1.6 kW with only a 25 cm gain length. The RotoCOIL device was based on the rotating disk singlet oxygen generator design as shown in Figure 12 and represents a third stage in COIL development emphasizing engineering demonstrations. By 1989, the supersonic RotoCOIL device was operated at 39 kW. Today, the Airborne Laser program employs a COIL device as the laser weapon.

4.2 Solid State Lasers

Solid-state lasers (SSL's) include heat-capacity, fibers and continuously cooled lasers. In the past, flash-lamp-pumped SSL's have produced high peak power, at the kW level, with good beam quality. However, it is the development of high efficiency, laser diode arrays that enables the possibility of higher average power SSL weapons. They require specialized thermal management systems.

The primary host crystal for SSL's has been yttrium aluminum garnet ($\text{Y}_3\text{Al}_5\text{O}_{12}$, YAG). YAG has attractive qualities of high thermal conductivity (0.14 W/cm-K), high mechanical strength, and good optical quality,¹² but a negative aspect is an inability to be grown to large sizes (i.e., < 2-cm diameters and 25-cm lengths for rods and < $0.7 \times 2.6 \times 21 \text{ cm}^3$ for slabs).¹³ For this size rod, energy storage is limited to < 50 J. However, under pulsed operation with pulse rates > 1 kHz, average powers of multi-kW can be achieved.

Optical transitions in these lasers occur on the energy levels of rare-earth ions doped into the crystal. Two common ions are neodymium (Nd^{3+}) and ytterbium (Yb^{3+}), which substitute into the lattice for Y^{3+} at ~0.2-1.4%. Energy diagrams of Nd:YAG and Yb:YAG are shown in Figure 14. For Nd:YAG, several pump bands exist at energies > $12 \times 10^3 \text{ cm}^{-1}$. Older Nd:YAG systems were flash-lamp pumped to fill these bands. Modern systems use semiconductor diode laser arrays at a wavelength of ~808 nm (typically with $\Delta\lambda \sim 3 \text{ nm}$) for much more efficient pumping. 70% flash lamp electrical efficiencies combined with ~1% optical-to-optical efficiency in filling these bands led to < 1% overall efficiencies. Diode efficiencies on the order of 50% now combine with ~20% optical-to-optical efficiencies to bring overall efficiencies up to ~10%.

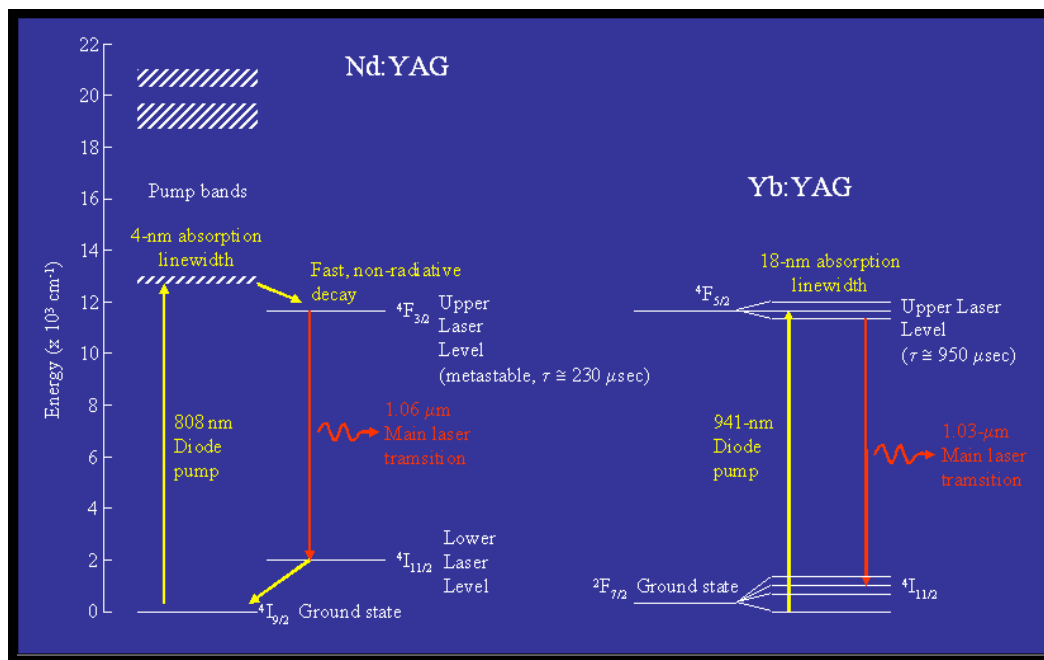


Figure 14. Energy diagrams for Nd:YAG and Yb:YAG.

Excited electrons decay nonradiatively into the upper laser level, $4\text{F}_{3/2}$, which is metastable with a radiative lifetime of ~230 μs . This allows for the build up of energy during Q-switched, pulsed operation, resulting in high peak-power pulses. The lasing transition occurs on the $4\text{F}_{3/2}$ and $4\text{I}_{11/2}$ levels, producing the 1.06- μm output.

The energy diagram for Yb:YAG shows three distinct advantages over the Nd:YAG system. First, the absorption band has an 18-nm line-width as opposed to 4 nm. This allows for reduced tolerance in the temperature control of the diode pump lasers. Second, the quantum defect between the 941-nm diode pump wavelength and the 1.03- μm Yb:YAG

lasing wavelength is less than the 808 nm to 1.06 μm in Nd:YAG, meaning higher quantum efficiencies and less waste heat. Third, the upper-state lifetime of Yb:YAG is 950 ms, allowing a reduced duty cycle. A disadvantage of the Yb:YAG system is that it is a quasi-three-level system, which means it has to be cooled to a greater extent than Nd:YAG, a four-level system, to keep the lower laser level depopulated.

Transverse mode control in a SS rod laser is achieved by the physical dimensions of the rod itself or by an inserted spatial filter. As expected, increasing the aperture size increases the output power, but also increases beam divergence and ultimately reduces brightness as higher order modes may be allowed to propagate. A larger problem in the rod devices is that the refractive index of YAG is temperature dependent ($n = 1.8169$ and $dn/dT = 8.9 \times 10^{-6} \text{ K}^{-1}$ at 1.064 μm) leading to thermal lensing problems. Recall that the optical-to-optical efficiencies for an Nd:YAG laser are on the order of 1% for flash lamp pumped devices and 20% for diode pumped devices due to, e.g., the quantum defect and a partial spatial overlap of the pump light and lasing mode in both the end-pumped and side-pumped configurations. The waste heat is extracted radially from the rod. A heat conduction analysis yields a steady-state temperature along the radius of a rod as

$$T(r) = T(r_0) + (Q/4K)(r_0^2 - r^2) \quad (12)$$

where r_0 is the radius of the rod, $T(r_0)$ is the temperature at the rod surface, Q is the heat generated per unit volume ($= P_a/(\pi r_0^2 L)$ where P_a is the total heat dissipated and L is the length of the rod), and K is the thermal conductivity of the rod. Figure 15 shows an example radial temperature distribution in a rod, which in turn, yields a thermal lens.

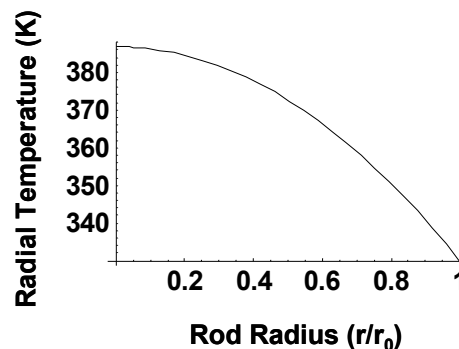


Figure 15. Temperature distribution in a YAG rod.

SSL's are used as the track illuminator (TILL) and beacon illuminator (BILL) on the Airborne Laser (ABL). The TILL is a Q-switched, 10-W-class, Yb:YAG laser comprised of four mutually incoherent illuminators sharing a single aperture. The return signal is integrated. The BILL is a kW-class Nd:YAG laser comprised of two laser modules which are multiplexed (i.e., share an aperture and are temporally interleaved) to provide pulses at 5 kHz.¹⁴

To alleviate the problems of thermal lensing in YAG rods, other cavity designs are being explored, particularly slabs and fibers. The advantage of the slab laser is depicted in Figure 16. Since heat is extracted over the entire face of the slab, temperature distribution across the face of the slab, i.e., transversely across the lasing mode, is uniform. The problem of small YAG crystals remains, so other host media are also explored, e.g., glass. Although the thermal problems may be reduced and larger slabs are possible, radiative lifetimes are reduced in this host and since the rare-earth ions no longer substitute on the Y lattice sites, the line-width is inhomogeneously broadened. Although energy storage in Nd^{3+} :glass is reduced to $< 500 \text{ kJ/m}^3$, with the increased size, output energy can be increased to $< 1 \text{ MJ}$, and peak powers to $< 10^{15} \text{ W}$.¹⁵

A laser cavity designed with solid-state slabs is utilized in the "heat capacity" laser. The heat capacity laser temporally separates cooling from lasing. The gain medium is spatially uniformly pumped by pulsed diode laser arrays.¹⁶ The pulse repetition rate is 5 ms, with a 10% duty cycle. The diode pumping, and thus lasing, occurs for approximately 10 s. As it does, waste heat is dumped to the slab and temperature increases. Post lasing, aggressive cooling occurs for $\sim 30 \text{ s}$ to 5 minutes to remove this heat, and the process may be repeated.¹⁵

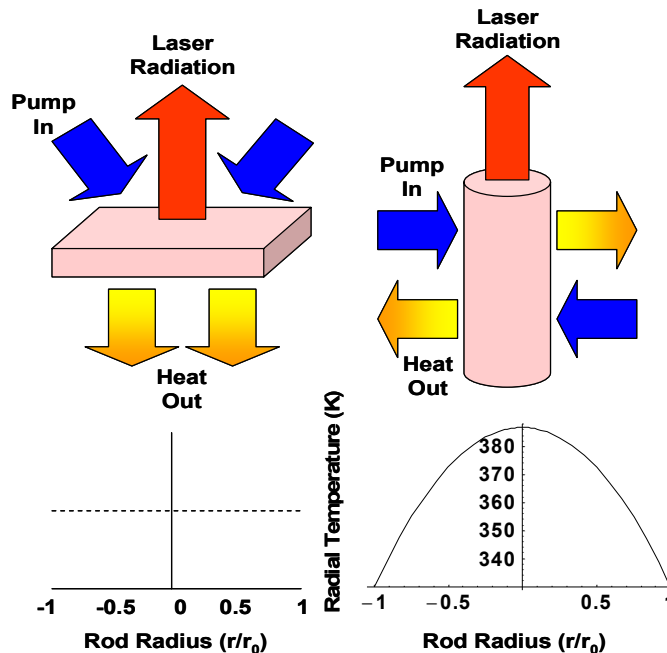


Figure 16. Thermal advantage of solid state slabs.

A flash-lamp-pumped Nd:glass heat capacity laser test-bed was developed at Lawrence Livermore National Laboratory (LLNL). The objectives of this test-bed were 500 J/pulse with a 20 Hz pulse rate for an average power of 10 kW with a run duration of 10 s and beam divergence < 40 mrad. These objectives were exceeded with 700 J/pulse demonstrated for an average power of 13 kW (and the other objectives met). LLNL also has a conceptual design for a 100-kW heat capacity laser. In this design, a new host crystal, gadolinium gallium garnet ($\text{Gd}_3\text{Ga}_5\text{O}_{12}$, “GGG”), is to be used. Again, the Nd^{3+} ions substitute into the lattice for Gd^{3+} at 0.3-1.5%. The thermal conductivity is about half that of YAG (0.07 W/cm-K), but thermal diffusivity is ten times that of glass and the fracture stress is five times that of glass. The benefit of GGG is that 20-cm-diameter crystal boules are able to be produced, yielding $5 \times 10 \text{ cm}^2$ slabs (as opposed to $0.7 \times 2.6 \text{ cm}^2$ maximum for YAG). The 100-kW design calls for nine $12 \times 12 \text{ cm}^2$ Nd:GGG slabs. Diode lasers will be used to produce 3.5 MW peak pump powers yielding 500 J/pulse output energies for 100-kW average powers. The projected optical-to-optical efficiency is 33% and wall-plug efficiency is 10%.¹⁵

Another design for a 100-kW SSL uses thin slab amplifiers in a reflective rather than transmissive mode. The advantage is that the backside is then available for liquid cooling, which is more efficient than gas cooling which is generally used for the transmissive designs. In the Boeing designs, the slab can be diode-pumped either from the back through an optically transparent rigid substrate or from the side. Scaling to high powers is achieved by increasing the number of amplifiers and by increasing the size of the amplifier itself.¹⁷

A third design for a 100-kW SSL uses the master oscillator-power amplifier (MOPA) approach. Here, a relatively low power laser (i.e., the MO) produces a high quality beam that is amplified by the PA's. In the Raytheon design for a phase-conjugate (PC)-MOPA a ~ 0.1 -1-kW, TEM_{00} , diffraction-limited Yb:YAG rod serves as the MO. Its output is coupled into a series of PA's which increase power by $\sim 10\times$, but reduce beam quality. The PA output is propagated into a loop PC mirror using a thermal liquid (acetone) as the nonlinear element. Amplifiers in this loop allow a PC reflectivity of ~ 3 with PC fidelity $> 50\%$. The conjugated beam is back-propagated through the PA's again amplifying by $\sim 10\times$, but now cleaning up the beam. Other MOPA designs could have the PA's in parallel rather than in series and would call for a coherent beam combining technique.¹⁸

Optical pumps for SSL's are arrays of semiconductor diode lasers. An individual laser may only have output power on the order of 1-5 W CW, but a monolithic bar ~ 1 -cm wide can have output powers > 250 W CW.¹⁹ Diode lasers are very

reliable with array lifetimes > 5,000 hours with minimal degradation.²⁰ The bars, then, can be stacked into modules producing > 100 kW of optical power with very high electrical-to-optical efficiency on the order of 50%. The Defense Advanced Research Projects Agency (DARPA) Super High Efficiency Diode Sources (SHEDS) program has goals of 80% efficient, 480-W diode stacks with $\Delta\lambda = 2 \pm 0.5$ nm for pumping SSL's.²¹

Thermal management of diode laser pumps is a significant technology. E.g., for 1-W emitters on 200- μ m centers producing a 1-cm-wide, 50-W array that is 50% efficient, waste heat extraction must be on the order of 1 kW/cm², and temperature control of a semiconductor laser to achieve wavelength drift < 3 nm must be ~ 10 °C. One method is rack-and-stack micro-channel technology which utilizes micro-channels on the order of 25 x 150 mm². LLNL has also developed silicon monolithic micro-channel (SiMM) technology with 30- μ m micro-channels on 60- μ m centers. Several (e.g., ten) diode bars make up a SiMM, which are then tiled together into large arrays.

The overall weight of a SSL system is dependent on the type of SSL (YAG, GGG or glass host; Nd or Yb lasing species; rod, slab, disc, or heat capacity design), pump diode efficiency (< 50% current, > 75% projected) and type of cooling (micro-channel, spray), application (ground- or space-based), power supply (conventional turbine generator, super-conducting generator, battery), and other parameters. Weight estimates typically range ~ 0.01 -0.5 lb/W. Estimates range from 500 Mg for a 10-MW_{avg} Yb:YAG space-based system with current battery technology (0.11 lb/W) to 44 Mg for the same system with turbo-alternator machinery (0.01 lb/W). Other estimates include 15 Mg for a 100-kW Nd:glass SSL with 45%-efficient diode pumps and micro-channel cooling (0.34 lb/W) to 1 Mg for a 100-kW Yb:YAG SSL with 75%-efficient diodes and spray cooling (0.023 lb/W).²²

4.3 Free Electron Lasers

The free electron laser (FEL) converts kinetic energy of an electron beam into coherent laser radiation by accelerating electrons to nearly the speed of light and sending a bunch of these electrons through a wiggler consisting of a series of magnets. The electrons bunch due to laser induced energy modulation, resulting in coherent radiation. The first FEL device was demonstrated using the Stanford superconducting accelerator in 1976.⁶ A schematic of the 1999 Jefferson Laboratory FEL with its photoinjected, superconducting rf linac is shown in Figure 17.

FEL's have two primary advantages as a source for laser weapons. First FEL's operate at a wide range of wavelengths, λ , specified by the resonance condition:

$$\lambda = \frac{\lambda_w}{2\gamma^2} (1 + a_w^2) \quad (13)$$

where the relativistic factor $\gamma^2 = (1 - v^2 / c^2)$ is specified by the speed of the electrons, v ; λ_w is the spatial period of the wiggler, and $a_w = e B \lambda_w / (2\pi m c^2)$ is determined by the magnetic field strength, B . Adjusting either the electron beam kinetic energy or the magnetic field strength tunes the wavelength easily and rapidly over a wide range. Secondly, very high peak powers can be achieved with no thermal lensing or other materials problems. The primary disadvantages of an FEL are size, weight and cost. The required electron beam energy increases at shorter wavelengths, increasing the size and cost. Most scientific uses of FELs have been in the infrared, although wavelengths in the ultraviolet have been demonstrated.

The following tables provide typical FEL operating parameters. The Jefferson Laboratory FEL achieved an average power of 1.72 kW in 1999.²³

Table 2. FEL operational parameters

Parameter (Units)	Demonstrated Parameter Range
Beam Current, I (Amps)	0.001 – 300
Beam Energy, E (MeV)	0.04 – 1000
Wavelength, λ (μ m)	0.1 - 200
Wiggler Period, λ_w (cm)	0.9 – 300
Wiggler Parameter, a_w	0.4 – 4.5

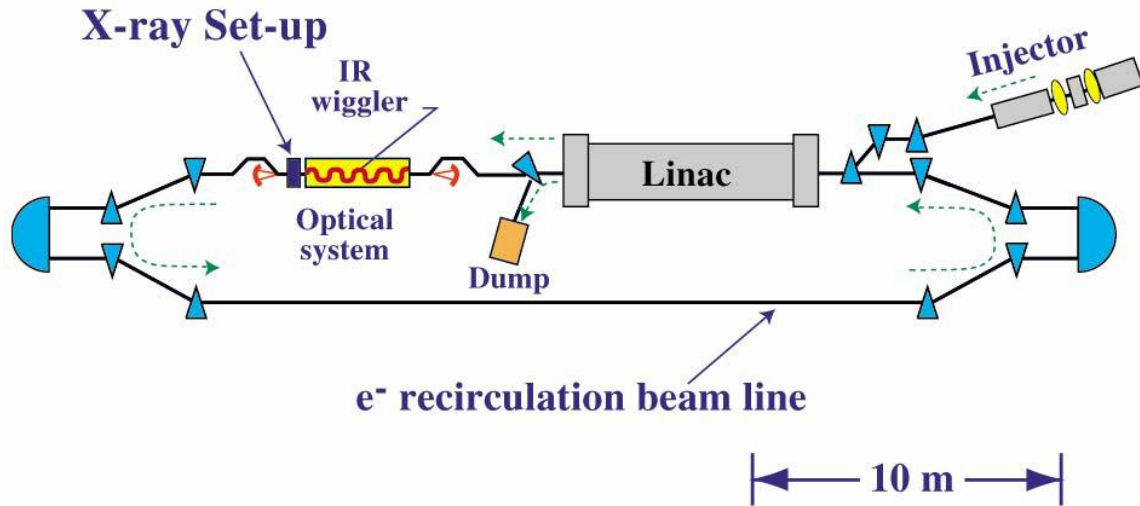


Figure 17. JLab linear accelerator, free electron laser.

Table 3. Jefferson Laboratory FEL of 1999²³

Parameter (Units)	Achieved Value
Average Power (kW)	1.72
Wavelength (μm)	3, 5, and 6
Energy / Pulse	< 70 μJ
Pulse Length (ps)	0.5 – 1.7
Repetition rate (MHz)	18.7 – 74.8
Beam Quality	2

5.0 THE BRIGHTNESS EQUATION

The ability for a high energy laser to damage a distant target depends on both the power of the laser device and the size of the minimum spot that can be delivered. Brightness, B , is a range independent figure of merit and corresponds to the radiometric quantity intensity (W/steradian):

$$B = \frac{\pi}{4} \left(\frac{P}{(\lambda/D)^2} \right) S \quad (14)$$

where P = laser power, λ = laser wavelength, D = diameter of the primary aperture, and S = total system Strehl. Diffraction at the primary aperture will limit the minimum angular divergence of the beam. The beam spot size may be further degraded by laser device beam quality, the beam control system, and most importantly, the atmosphere. These effects are often summarized by a system Strehl, S , defined as the ratio of the on axis target intensity actually achieved to the on axis intensity limited only by diffraction. A schematic of the brightness equation for a typical airborne application is provided in Figure 18. A 30 cm spot at target distance of 300 km yields a solid angle of $\Omega \sim 10^{-12}$ str, or a one-dimensional angle of $\Theta \sim 1$ μradian. For a laser device power of $P \sim 1$ MW, the resulting brightness would be, $B \sim 10^{18}$ W/str.

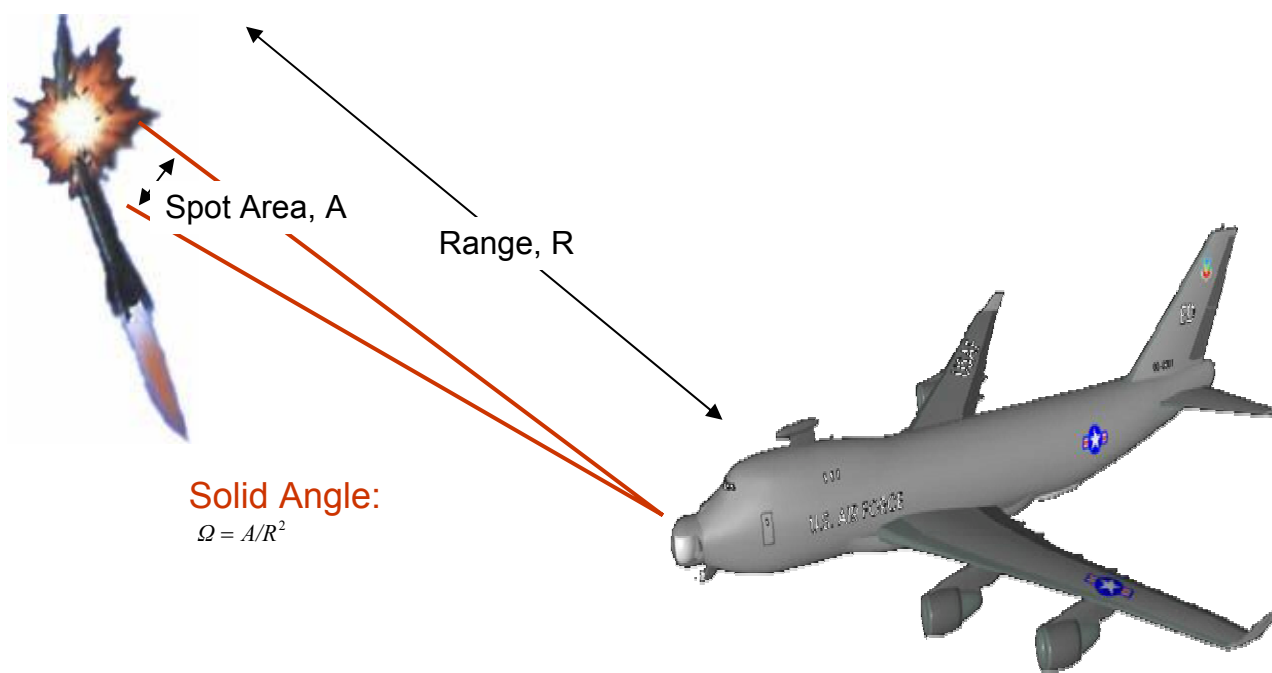


Figure 18. Brightness as a figure of merit.

6.0 BEAM CONTROL AND ATMOSPHERIC EFFECTS

6.1 Atmospheric Effects

There are numerous effects that need to be accounted for to understand what happens to a laser beam as it is propagated through the atmosphere. These are grouped into four areas, and summarized in Figure 19 to facilitate discussion.

Diffraction is an unavoidable consequence of the nature of light and will tend to spread out the energy as the beam propagates, even through a perfect vacuum. This implies that there is a fundamental limit to the smallest size of a spot that a laser can be focused down to. This amount of spreading is proportional to the ratio λ/D , where D is aperture diameter and λ is wavelength of light. Thus for larger apertures or smaller wavelengths, the effect of diffraction is reduced.

Scattering is due to atmospheric constituents that do not perfectly transmit the beam, but alter its path. The result is a beam that is spread beyond what would be otherwise expected. The best way to counteract scattering is to reduce the particulates in the atmosphere. However, this is rarely possible, so longer wavelengths tend to reduce the effect of scattering.

Absorption will decrease the power that is transmitted to the target by removing energy from the beam as it propagates. Different atmospheric constituents will absorb different wavelengths at various levels, which leads to a very strong dependence on location, time of year and weather patterns. Especially in the maritime environment, water vapor absorption will dominate. As an additional note, it is absorption that causes the atmosphere to heat up. When this occurs at a high enough rate, thermal blooming occurs.

In addition to diffraction, the atmospheric turbulence will cause the beam to spread and reduce power delivered to the target. The strength of the turbulence is measured with the parameter, C_n^2 . This will be highly dependent on altitude as well as wind speed and other atmospheric parameters. The primary effect will be beam spread, but the turbulence will also induce a random beam walk around the aim point.

Propagation Effects

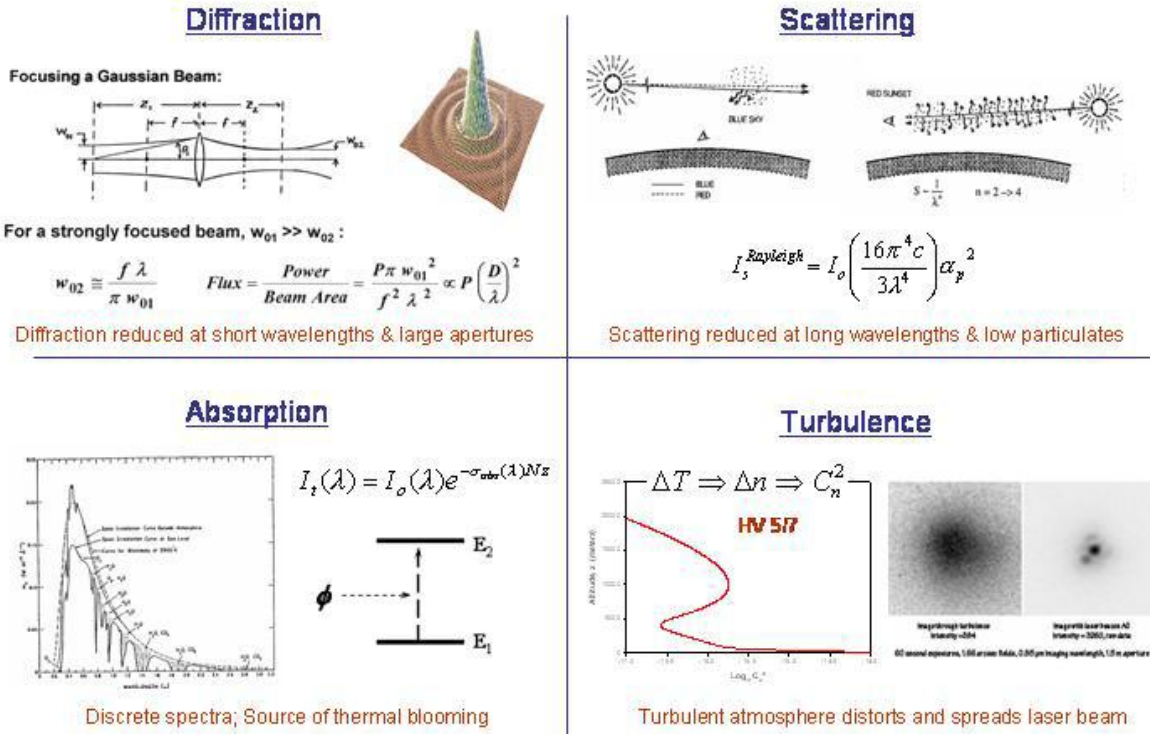


Figure 19. Atmospheric propagation effects.

In order to provide a more intuitive and quantitative understanding of turbulence, several parameters are often used. The first is the Fried coherence length, also called r_0 . Small values of r_0 correspond to strong turbulence and poor seeing, while large values mean weak turbulence and good seeing. The value of r_0 specifies the distance over which the effects of the atmosphere are small (mathematically, the mean-square-deviation of the phase is 1 rad^2). Thus, if the laser beam we are transmitting is about the same size as r_0 , the effects of turbulence will be predominantly only a beam wander. The Fried coherence length is computed from the C_n^2 altitude profile:

$$r_0 = \left[\frac{2.905}{6.88} \left(\frac{2\pi}{\lambda} \right)^2 \int_{\text{Path}} C_n^2(z) \left(\frac{R-z}{R} \right)^{5/3} dz \right]^{-3/5} \quad (15)$$

Typical values are $r_0 = 1 - 10 \text{ cm}$ for tactical scenario and $r_0 = 0.1 - 1 \text{ m}$ for strategic scenarios and vary greatly based on target range and current meteorological conditions.

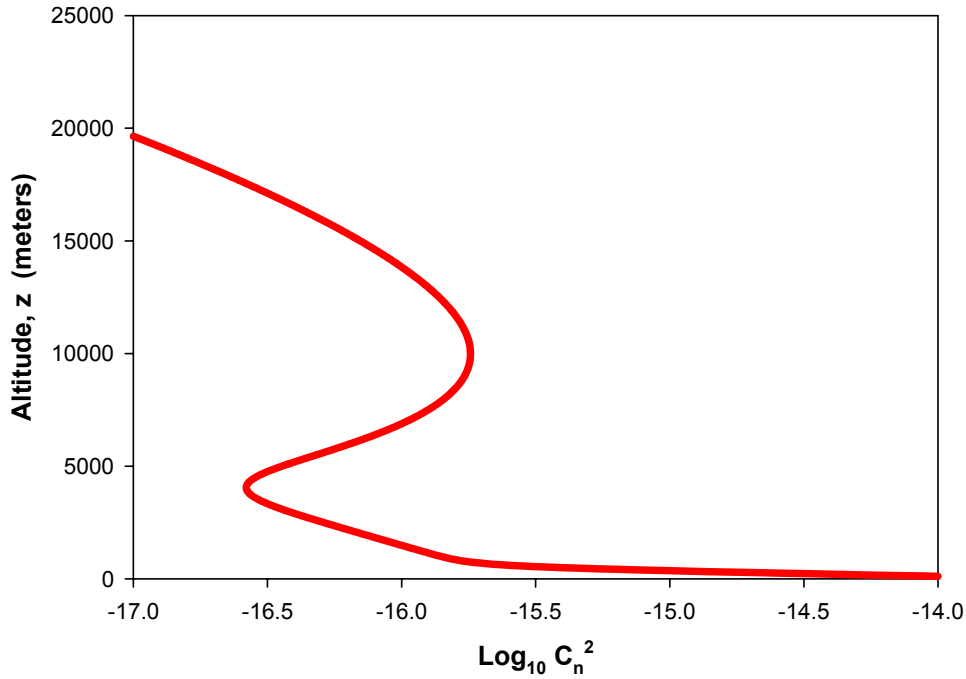


Figure 20. A typical (HV 5/7) optical turbulence altitude profile.

The Isoplanatic angle is another parameter used to characterize the turbulence. It measures the maximum angular offset between two beams where the effect of turbulence on both beams is nearly identical. This is important for cases where we measure the atmospheric turbulence with a beacon that may be offset from the aimpoint. The isoplanatic angle specifies the maximum offset before performance degrades significantly.

$$\theta_0 = \left[2.905 \left(\frac{2\pi}{\lambda} \right)^2 \int_{Path} C_n^2(z) z^{5/3} dz \right]^{-3/5} \quad (16)$$

If the isoplanatic angle is smaller than the diffraction angle, $\theta_0 < \lambda/D$, the adaptive optics methods discussed below will be largely ineffective.

Finally the Greenwood frequency is a measure of how fast the atmosphere is changing and specifies the bandwidth required of the adaptive optics control system. Large values typically indicate strong turbulence. Typical values are $f_G < 100$ Hz when the target is slow and close to the target and $f_G \sim 1000$ Hz for fast distant targets.

$$f_G = \left[0.102 \left(\frac{2\pi}{\lambda} \right)^2 \int_{Path} C_n^2(z) W_E^{5/3}(z) dz \right]^{3/5} \quad (17)$$

Thermal blooming is another source of laser beam aberration arising from atmospheric absorption. Absorption causes the temperature of the air that the laser beam is passing through to increase. The typical result is a crescent shaped beam spot. The amount of thermal blooming is typically characterized by a single number, the thermal distortion parameter.

$$N_D = \frac{C_A \cdot k \cdot P}{D} \int_{Path} F(r) \cdot \frac{\alpha_{abs}(r) \cdot T_0 \cdot \tau(r)}{V_E(r) \cdot T(r)} dr \quad (18)$$

This parameter represents the number of waves of optical path length difference induced by the temperature gradient. Note that the temperature increases for thermal blooming are quite small, $\Delta T < 0.001$ C, particularly in comparison to similar values for turbulence of 0.1 C. However, the temperature effects are cumulative in the case of thermal blooming, whereas the effects are random for turbulence. Tactical missions where the targets are nearly stationary, the atmosphere is thicker at low altitude leads to higher thermal blooming. For very high thermal blooming conditions, the power delivered to the target actually decreases as the source power increases, leading to a critical maximum power.

The irradiance pattern for thermal blooming with a distortion number of $N_D = 1$ is shown in Figure 21.

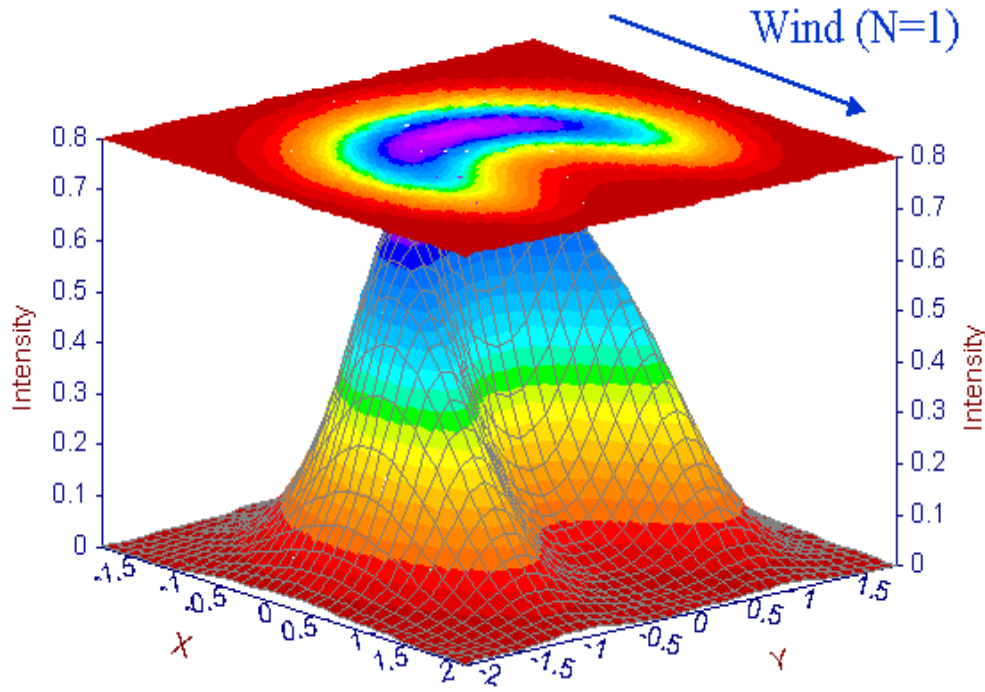


Figure 21. Typical irradiance pattern for thermal blooming with $N_D = 1$.

Adaptive optics (AO) may be employed to partially reduce the detrimental effects of atmospheric propagation. An AO system usually include a beacon laser to probe the state of the atmosphere, a wavefront sensor to analyze the return signal from the beacon, a deformable mirror capable of correcting the wavefront on the outgoing laser beam, and a high speed control system.

The deformable mirror is typically small, ~ 10 cm, and is located near the image of a pupil. The number of actuators to deform the mirror is typically 10-1000, and is based on a trade-off between the degree of correction required and the faint return signal from the beacon.

A large number of actuators require a similarly large number of sub-apertures in the wavefront sensor. The sub-aperture size must be on the order of the Fried coherence length.

Table 4 . Elements of Beam Control

Element	Parameter	Value	Scale / Effect
Primary Aperture	Diameter D	$D = 0.1 - 10$ meters	λ/D Diffraction
Fast Steering Mirror	Jitter ϕ, j	$\phi = 10 - 200$ nradian	$j / (\lambda/D) = 0.2 - 1.0$ Blurred Beam
Deformable Mirror	Number of actuators, N	$N = 10^1 - 10^3$	High order AO correction
Wavefront Sensor	Sub-aperture size, d	$d / r_0 \sim 2$ (r_0 = Fried coherence length)	$N \sim (D/d)^2$ Fitting Error
Beacon / Illuminator	Beam radius at target, a	$(a/R) / \theta_0 < 1$ (θ_0 = beam radius angle, R = Range)	Sense Turbulence Track Target
Control System	AO Bandwidth, f	$(f / f_G) > 2$ (f_G = Greenwood Frequency)	Temporal fluctuations

3. TARGET LETHALITY AND WEAPON EFFECTIVENESS

Lethality is the capability of a weapons system to render a target non-functional. Targets may be destroyed, denied, degraded or delayed. Kill assessment is complicated if the target is not destroyed.

Lethal fluence is defined as the energy per unit area of illuminated target required for damage. Typical values are 1- 10 kJ/cm² and depend on target materials, particularly across classes of materials such as metals, organics, refractory materials, and coatings. Some targets also require the fluence to be delivered over a minimum area. Irradiance, I, is the energy flux delivered to the target so that the fluence, F, accumulated of a dwell time, τ , is

$$F = I \tau \quad (19)$$

Irradiances of about 1 kW/cm² require dwell times of 1-10 seconds to meet the lethal fluence.

Weapon effectiveness extends beyond lethality to include fielding a functioning weapon system. The weaponeering process assessing the relative effectiveness of various weapon systems for the attack of a given target. Issues of operability, supportability, and cost must be included in such considerations. Today, lasers remain an emerging technology and represent higher risk solutions that existing conventional weapons. Laser weapons may find their first application of the battlefield in niche, high priority missions where no viable alternatives exist.

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